SIVAKORN KHUNDILOKRATTAYA

Computer Engineering Student • Aspiring Data Scientist & ML Engineer
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EDUCATION

Srinakharinwirot University (SWU) 2024 – Present (Expected 2026)
Bachelor of Engineering in Computer Engineering *Bangkok, Thailand*

- Specialization in Statistical Machine Learning, Financial Risk Modeling, and Distributed Data Engineering.

TECHNICAL SKILLS & COMPETENCIES

Languages: Python (Expert), SQL (PostgreSQL, MySQL), C, C++, Java, R, Bash scripting.

ML & Explainable AI: Scikit-Learn, LightGBM, XGBoost, PyTorch Geometric (R-GCN), ONNX Runtime, SHAP (TreeExplainer), LIME, OptBinning, SMOTE-Tomek, Pandas, NumPy.

Data Engineering & Tools: PostgreSQL, MySQL, Apache Spark, Apache Airflow, Docker, FastAPI, Git, CI/CD pipelines.

Spoken Languages: Thai (Native), English (B1+ Professional), Japanese (Elementary).

FEATURED ENGINEERING PROJECTS

Enterprise Credit Risk Scoring & Explainable AI (XAI) 2026
Python, LightGBM, SHAP, FastAPI, Docker • Basel II/III Credit Decisioning Engine

- Engineered Weight of Evidence (WoE) and Information Value (IV) transformation pipeline to discretize continuous features into monotonic risk bins.
- Integrated SHAP TreeExplainer to compute exact local Shapley values, automatically generating legally defensible Adverse Action notices (ECOA/FCRA compliant).
- Deployed an asynchronous containerized FastAPI microservice delivering Probability of Default (PD) scoring with sub-50ms inference latency.

FlowSense & TrustGraph – Autonomous Cashflow & Graph Fraud Defense 2026
PyTorch Geometric (R-GCN), LightGBM, ONNX Runtime, FastAPI, React 18 • KBTG Kampus Hackathon 2026

- Engineered FlowSense 30-day liquidity forecasting using LightGBM with rolling lag features, normal-dip alert suppression, and surplus-only micro-savings sweeps with 1-tap recall.
- Built TrustGraph pre-transfer fraud defense using Relational Graph Convolutional Networks (R-GCN) to embed account topology, executing sub-12ms P99 inference via ONNX Runtime (<80ms SLA).
- Formulated target-specific explainable risk factors (e.g., dormant account reactivation, transaction velocity surges) paired with a 5-second facial liveness micro-authentication gateway.

Shopee Smart COD Reliability & Dynamic Decisioning Engine 2026
Python, Scikit-Learn, Random Forest, FastAPI, Vanilla JS • Shopee Ultimate Case Challenge (SUCC 2026)

- Developed calibrated Random Forest ML pipeline (0.8629 ROC-AUC, 0.3722 PR-AUC, 0.0208 Brier score) across 50,000 synthetic orders / 12,000 buyer profiles to predict COD failure probabilities.
- Statistically verified 10.6× COD failure risk concentration via two-proportion Z-tests ($z=24.42$, $p < 0.001$), translating failure probabilities into an adaptive 0–100 buyer Reliability Score.
- Architected 4-tier checkout policy (Grade A 1-click COD to Grade D 30% deposit) with dynamic feedback loops (+8/-20 pts) and EasySell fraud checks on an interactive smartphone simulator.

Long Overdue Debtor (LOD) Classification System Dec 2025
LightGBM, XGBoost, Scikit-Learn • Aihack Thailand 2025 National Finalist

- Formulated temporal delinquency velocity ratios and revolving limit exhaustion slopes across 32,524 train / 8,619 test blind split samples.
- Conducted 5-Fold Stratified Cross-Validation on tuned LightGBM and XGBoost ensembles, achieving 0.894 ROC-AUC and placing in the National Finalist tier.

COMPETITIVE HACKATHONS & INDUSTRY CHALLENGES

KBTG Kampus Hackathon 2026 – Track 2 Lead (Data Science & Intelligence) 2026
Kasikorn Business-Technology Group (KBTG) • Status: Advanced to Next Round *Bangkok, Thailand*

- Spearheaded data intelligence architecture for K PLUS, connecting FlowSense cashflow projection and TrustGraph relational fraud screening.
- Designed R-GCN graph neural network embeddings with ONNX Runtime sub-12ms P99 inference, modeling projected 1.2B–2.0B THB in CASA deposit mobilization and >85% fraud interdiction.

Shopee Ultimate Case Challenge (SUCC) 2026 – Lead Data Scientist & Product Architect 2026
Shopee Thailand Operations & Logistics Strategy Track *Bangkok, Thailand*

- Designed end-to-end data science and decisioning platform to reduce failed Cash-on-Delivery (COD) orders and reverse logistics return-to-origin (RTO) waste.
- Built full-stack interactive smartphone shopping simulator and FastAPI portal modeling 3.58M THB in potential monthly savings through 4-tier risk grading and delivery window scheduling.

AIRA & AIFUL / Aihack Thailand 2025 National Finalist Dec 2025
Chulalongkorn Business School × ProbSpace *Bangkok, Thailand*

- **Role & Contribution:** Machine Learning & Financial Risk Modeling Lead on ~40,000 real-world consumer loan samples.
- **Technical Rigor:** Built leak-free temporal feature matrices, handling class imbalance (1:12 default ratio) with SMOTE-Tomek and target encoding (0.894 ROC-AUC).
- **Executive Pitch:** Solved Question Sheet balancing default risk reduction against interest margin profitability, pitching to C-level executives.

True Innovation Launchpad 2026 – AI & Data Science Lead Jan 2026
True Corporation Healthcare Innovation Track (Team Nakphatthana Tuapralat) *Bangkok, Thailand*

- Spearheaded data strategy and predictive modeling for elderly frailty prevention, developing sub-30ms clinical risk scoring APIs (GuardianAI).

GISTDA Spatial Resilience Hackathon & LINE MAN Case Competition 2024 – 2026
Spatial Lead (GISTDA Rain-to-Flood 30m Mapping) • Growth Lead (LINE MAN Wongnai 8.87 THB CAC) *Bangkok, Thailand*

- Built "Rain-to-Flood" spatial model fusing Sentinel-1 SAR, DEM, and precipitation APIs; formulated LINE MAN campus unit economics campaign acquiring 462 VIP trial users.

INDUSTRY CERTIFICATIONS & STANDARDIZED ASSESSMENTS

Data Analyst Assessment (STAMP) – Score: 97/150 (Intermediate) March 2026
SKILLKAMP by KBTG (Kasikorn Business-Technology Group) • Benchmark: 90

- **Core Scores:** Data Cleaning & Preprocessing: 23.00/30 | EDA & Viz: 20.00/30 | ML for Analysts: 18.00/30 | Foundations: 20.00/30

Digital and Performance Marketer Assessment (STAMP) – Score: 98/150 (Intermediate) March 2026
SKILLKAMP by KBTG • Benchmark: 94

- **Top Performer:** Digital Marketing Analytics: 26.00/30 (Benchmark: 17.94) | Data-Driven Marketing: 21.00/30 | Content: 19.00/30.

Oxford Placement Test – Score: 51/120 (CEFR B1+ Professional) June 2026
AUA Language Center / Oxford University Press

- Grammar & Use of English: 50.88 (B1) | Listening Proficiency: 51.12 (B1) – Verified professional working proficiency in English.

COURSEWORK & SOFTWARE ENGINEERING DISCIPLINE

Core Coursework: Machine Learning, Database Systems, Operating Systems, Statistical Methods for Engineers, Software Engineering, Data Structures & Algorithms, Computer Networks.
Engineering Discipline: Adheres to PEP-8 style standards, modular OOP pipeline design, reproducible containerized Docker environments, and zero black-box deployments without local feature attribution.